

Normal Subgroups and Factor Groups

Road map

The set G/H of left cosets exists for every subgroup $H \leq G$, but coset multiplication

$$(aH)(bH) := (ab)H$$

is independent of representatives only for special subgroups. These are the normal subgroups. We prove the equivalence between normality, equality of left and right cosets, and invariance under conjugation; build the factor group G/N ; and show that kernels are normal. We then study canonical projections, centers, simple groups, the classification picture for finite simple groups, cyclic simple groups, A_4 , and the quotient S_n/A_n .

7.1 Why coset multiplication needs a condition

Recall that for a group G and a subgroup $H \leq G$,

$$G/H = \{aH : a \in G\}$$

is the set of all left cosets. A coset aH is generally not a subgroup: it contains the identity only when $a \in H$, in which case $aH = H$.

A natural attempt to multiply cosets is

$$aH \circ bH := (ab)H.$$

If this operation is well defined, its group axioms would follow from those of G :

- the identity would be $H = eH$, because

$$aH \circ H = (ae)H = aH \quad \text{and} \quad H \circ aH = (ea)H = aH;$$

- the inverse of aH would be $a^{-1}H$, because

$$aH \circ a^{-1}H = (aa^{-1})H = H;$$

- associativity would follow from

$$\begin{aligned} aH \circ (bH \circ cH) &= aH \circ (bc)H = (a(bc))H, \\ (aH \circ bH) \circ cH &= (ab)H \circ cH = ((ab)c)H, \end{aligned}$$

and $a(bc) = (ab)c$.

The real issue is *well-definedness*. A coset has many representatives. If

$$aH = a'H, \quad bH = b'H,$$

must it follow that

$$(ab)H = (a'b')H?$$

For an arbitrary subgroup, the answer can be no.

7.2 Normal subgroups

Definition 7.1: Normal subgroup

A subgroup $N \leq G$ is *normal* in G , written

$$N \trianglelefteq G,$$

if

$$aN = Na \quad \text{for every } a \in G.$$

Thus every left coset of N is the corresponding right coset.

Theorem 7.1: Equivalent forms of normality

Let $N \leq G$. The following are equivalent:

1. $N \trianglelefteq G$;
2. $gN = Ng$ for every $g \in G$;
3. $gNg^{-1} = N$ for every $g \in G$;
4. $gNg^{-1} \subseteq N$ for every $g \in G$.

证明. The first two statements are the same by definition. If $gN = Ng$, multiply both sets on the right by g^{-1} :

$$gNg^{-1} = Ng g^{-1} = N.$$

Conversely, if $gNg^{-1} = N$, multiply on the right by g to obtain $gN = Ng$. Thus (2) and (3) are equivalent.

Clearly (3) implies (4). Suppose (4) holds. Applying it to g^{-1} gives

$$g^{-1}Ng \subseteq N.$$

Conjugating this inclusion by g gives

$$N \subseteq gNg^{-1}.$$

Together with $gNg^{-1} \subseteq N$, this yields equality. Hence (4) implies (3). ■

Geometric meaning

Conjugation by g sends the subgroup N to the conjugate subgroup gNg^{-1} . Normality means that this operation never produces a new subgroup:

$$gNg^{-1} = N \quad \text{for all } g \in G.$$

The subgroup is invariant under all inner automorphisms.

7.3 Normality is exactly what makes the product well defined

Theorem 7.2: Well-defined coset multiplication

Let $N \leq G$. The rule

$$(aN)(bN) := (ab)N$$

defines a well-defined binary operation on G/N if and only if $N \trianglelefteq G$.

证明. Suppose first that $N \trianglelefteq G$. Let

$$aN = a'N, \quad bN = b'N.$$

By the coset-equality criterion, there are $n_1, n_2 \in N$ such that

$$a' = an_1, \quad b' = bn_2.$$

Then

$$\begin{aligned} a'b' &= (an_1)(bn_2) \\ &= ab(b^{-1}n_1b)n_2. \end{aligned}$$

Normality gives $b^{-1}n_1b \in N$, and closure of N gives

$$(b^{-1}n_1b)n_2 \in N.$$

Therefore $a'b' \in abN$, which is equivalent to

$$(a'b')N = (ab)N.$$

The product is independent of the chosen representatives.

Conversely, suppose the product is well defined. Fix $n \in N$ and $b \in G$. Since $nN = N = eN$, well-definedness gives

$$(nN)(bN) = (eN)(bN).$$

Therefore

$$nbN = bN,$$

so the left-coset criterion gives

$$b^{-1}nb \in N.$$

Since n was arbitrary,

$$b^{-1}Nb \subseteq N \quad \text{for every } b \in G.$$

The equivalent-normality theorem now shows that $N \trianglelefteq G$. ■

This calculation is the detailed version of the chain recorded in the handwritten notes. Starting from arbitrary representatives and applying the coset equality lemma eventually forces

$$Hb \subseteq bH \quad \text{and} \quad bH \subseteq Hb,$$

and hence $bH = Hb$ for every $b \in G$.

7.4 Factor groups

Definition 7.2: Factor group or quotient group

Let $N \trianglelefteq G$. The set

$$G/N = \{aN : a \in G\},$$

equipped with

$$aN \cdot bN := (ab)N,$$

is called the *factor group* or *quotient group* of G by N .

Theorem 7.3

If $N \trianglelefteq G$, then G/N is a group. Its identity is N , and

$$(aN)^{-1} = a^{-1}N.$$

证明. The preceding theorem proves that multiplication is well defined. Closure is automatic because $(ab)N \in G/N$. The calculations at the beginning of the chapter verify associativity, the identity N , and the inverse $a^{-1}N$. Thus all group axioms hold. ■

If G is finite, then

$$|G/N| = [G : N] = \frac{|G|}{|N|}.$$

Here $|G/N|$ counts cosets, not the number of elements inside one coset.

7.5 Kernels and images

Theorem 7.4: Kernels are normal

Let $\varphi : G \rightarrow G'$ be a homomorphism. Then

$$\ker \varphi \trianglelefteq G.$$

If $N \trianglelefteq G$, then

$$\varphi(N) \trianglelefteq \varphi(G) \leq G'.$$

In particular, if φ is surjective, then $\varphi(N) \trianglelefteq G'$.

证明. We already know that $\ker \varphi \leq G$. Let $a \in G$ and $x \in \ker \varphi$. Then

$$\begin{aligned} \varphi(axa^{-1}) &= \varphi(a)\varphi(x)\varphi(a)^{-1} \\ &= \varphi(a)e'\varphi(a)^{-1} = e'. \end{aligned}$$

Hence $axa^{-1} \in \ker \varphi$, and therefore

$$a(\ker \varphi)a^{-1} \subseteq \ker \varphi.$$

The normality criterion gives $\ker \varphi \trianglelefteq G$.

Now suppose $N \trianglelefteq G$. The image theorem gives $\varphi(N) \leq \varphi(G)$. Take $a' \in \varphi(G)$ and $x' \in \varphi(N)$. Choose $a \in G$ and $x \in N$ with

$$a' = \varphi(a), \quad x' = \varphi(x).$$

Then

$$\begin{aligned} a'x'(a')^{-1} &= \varphi(a)\varphi(x)\varphi(a)^{-1} \\ &= \varphi(axa^{-1}). \end{aligned}$$

Since $N \trianglelefteq G$, we have $axa^{-1} \in N$, so the last element belongs to $\varphi(N)$. Therefore

$$a'\varphi(N)(a')^{-1} \subseteq \varphi(N) \quad (a' \in \varphi(G)),$$

and $\varphi(N) \trianglelefteq \varphi(G)$. ■

Proposition 7.1: Subgroups of Abelian groups

If G is Abelian, every subgroup $N \leq G$ is normal.

证明. For $a \in G$ and $n \in N$,

$$ana^{-1} = naa^{-1} = n \in N.$$

Thus $aNa^{-1} = N$ for every a , and $N \trianglelefteq G$. ■

For example,

$$m\mathbb{Z} \trianglelefteq \mathbb{Z}, \quad \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}_m.$$

7.6 The canonical projection

Let $N \trianglelefteq G$. Define

$$\pi : G \longrightarrow G/N, \quad a \longmapsto aN.$$

This is the *canonical projection* or *quotient map*. It is a homomorphism because

$$\begin{aligned}\pi(a)\pi(b) &= (aN)(bN) \\ &= (ab)N \\ &= \pi(ab).\end{aligned}$$

It is surjective by the definition of G/N . Its kernel is

$$\begin{aligned}\ker \pi &= \{a \in G : \pi(a) = N\} \\ &= \{a \in G : aN = N\} \\ &= N.\end{aligned}$$

This is a consistency check on the theory: every normal subgroup occurs as the kernel of its canonical projection.

Proposition 7.2: The center is normal

For every group G ,

$$Z(G) \trianglelefteq G.$$

Moreover, $Z(G)$ is Abelian.

证明. If $z \in Z(G)$ and $a \in G$, then $az = za$, so

$$aza^{-1} = zaa^{-1} = z.$$

Thus $aZ(G)a^{-1} = Z(G)$, proving normality. If $x, y \in Z(G)$, then x commutes with every element of G , in particular with y ; hence $xy = yx$. Therefore $Z(G)$ is Abelian. ■

7.7 Simple groups

Definition 7.3: Simple group

A nontrivial group G is called *simple* if its only normal subgroups are

$$\{e\} \quad \text{and} \quad G.$$

Simple groups are the indivisible building blocks of finite group theory: they have no nontrivial proper normal subgroup from which a smaller quotient can be formed.

7.7.1 The classification picture for finite simple groups

The handwritten lecture presents the classification of finite simple groups as a major milestone in mathematics. Its main branches are shown in Figure 7.1.

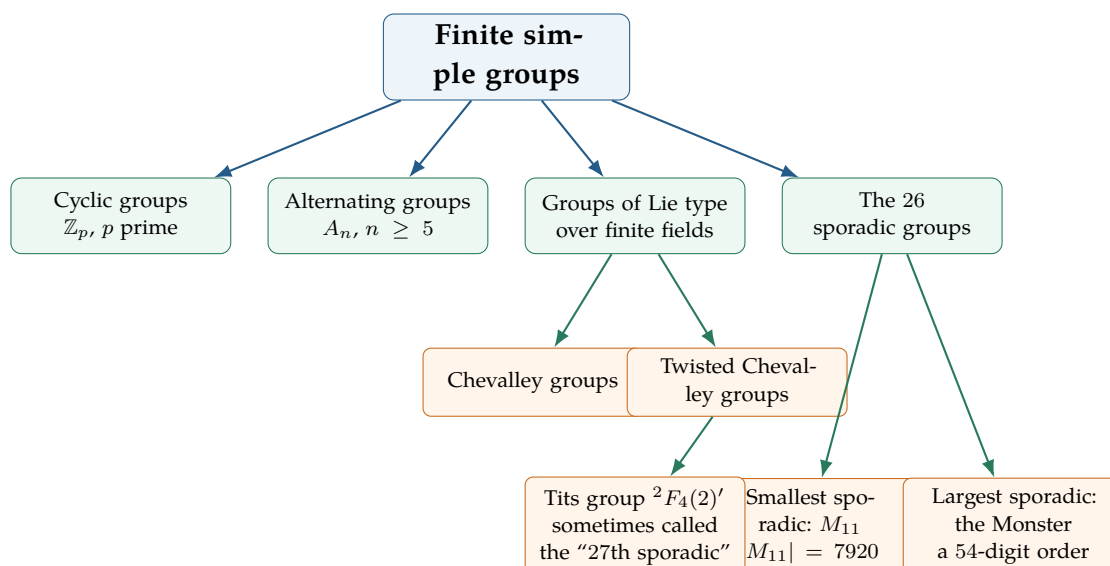


图 7.1: The classification tree recorded in the notes: three infinite families and 26 sporadic exceptions. The Tits group belongs to the Lie-type side but is historically nicknamed the “27th sporadic group.”

The details written around the original diagram include the following.

- The cyclic simple groups form the series $\mathbb{Z}_p$ for primes p .
- The alternating groups A_n are simple for $n \geq 5$.
- The groups of Lie type include Chevalley groups and twisted Chevalley groups constructed over finite fields.
- The smallest sporadic simple group is the Mathieu group M_{11} , with

$$|M_{11}| = 2^4 \cdot 3^2 \cdot 5 \cdot 11 = 7920.$$

- The largest sporadic group is the Monster, also called the Fischer–Griess Monster or the “Friendly Giant.” Its order is

$$|\mathbb{M}| = 808017424794512875886459904961710757005754368 \\ 000000000,$$

a 54-digit integer. The notes describe it through rotations in a 196883-dimensional space; more precisely, 196883 is the dimension of its smallest nontrivial complex irreducible representation.

- The Tits group ${}^2F_4(2)'$ has order

$$17\,971\,200 \approx 1.8 \times 10^7.$$

It is a finite simple group of Lie type, although it is often informally called the “27th sporadic group.”

- The handwritten note summarizes the classification proof as a collaboration of hundreds of authors over roughly 1955–2004, spread over thousands of pages.

For a field $\mathbb{F}$, the characteristic mentioned in the notes is

$$\text{char } \mathbb{F} := \min\{p \in \mathbb{N} : \underbrace{1 + \cdots + 1}_{p \text{ terms}} = 0\},$$

when such a positive integer exists; otherwise the characteristic is 0. Equivalently, for positive characteristic,

$$\text{char } \mathbb{F} = |\langle 1 \rangle_+|,$$

the order of the cyclic subgroup generated by 1 in the *additive* group of the field.

7.7.2 Exactly which cyclic groups are simple?

Theorem 7.5

The finite cyclic group $\mathbb{Z}_n$ is simple if and only if n is prime.

证明. If n is prime, Lagrange's theorem shows that a subgroup of $\mathbb{Z}_n$ has order 1 or n . Hence the only subgroups, and therefore the only normal subgroups, are $\{0\}$ and $\mathbb{Z}_n$.

Conversely, suppose n is composite. Choose a proper divisor k with

$$1 < k < n, \quad n = kr.$$

Let a be a generator of $\mathbb{Z}_n$ in multiplicative notation and put $b = a^k$. Then

$$b^r = (a^k)^r = a^{kr} = a^n = e.$$

The subgroup

$$\langle b \rangle = \{e, b, b^2, \dots, b^{r-1}\}$$

is nontrivial and proper. Since $\mathbb{Z}_n$ is Abelian, every subgroup is normal, so

$$\{e\} < \langle b \rangle < \mathbb{Z}_n \quad \text{and} \quad \langle b \rangle \trianglelefteq \mathbb{Z}_n.$$

Therefore $\mathbb{Z}_n$ is not simple. ■

Example 7.1: A_4 is not simple

Inside A_4 , consider the Klein four-subgroup

$$V_4 = \{e, (12)(34), (13)(24), (14)(23)\}.$$

This is the subgroup displayed in the handwritten notes in one-line notation:

$$\{(1234), (2143), (3412), (4321)\}.$$

For every $\sigma \in A_4$,

$$\sigma(ij)(k\ell)\sigma^{-1} = (\sigma(i) \sigma(j))(\sigma(k) \sigma(\ell)).$$

Thus conjugation permutes the three double transpositions and fixes e . Hence

$$\sigma V_4 \sigma^{-1} = V_4 \quad (\sigma \in A_4),$$

so $V_4 \trianglelefteq A_4$. Because

$$\{e\} < V_4 < A_4,$$

the group A_4 is not simple. This contrasts with the theorem that A_n is simple for every $n \geq 5$.

7.8 The quotient S_n/A_n

The sign homomorphism satisfies

$$A_n = \ker(\text{sgn}),$$

so the kernel theorem gives

$$A_n \trianglelefteq S_n.$$

For $n \geq 2$, let τ be any transposition. The two cosets are

$$S_n/A_n = \{A_n, \tau A_n\}.$$

Their multiplication is

$\cdot$	A_n	τA_n
A_n	A_n	τA_n
τA_n	τA_n	A_n

because $\tau^2 = e$. This is the Cayley table of $\mathbb{Z}_2$, and hence

$$S_n/A_n \cong \mathbb{Z}_2.$$

A quotient group is not usually a subgroup

The elements of S_n/A_n are the two sets A_n and τA_n ; they are not individual permutations. Although the quotient is abstractly isomorphic to the two-element subgroup $\{e, \tau\}$, that chosen copy need not be normal in S_n .

For example, in S_3 , let

$$\tau = (12), \quad \sigma = (123), \quad H = \{e, \tau\}.$$

Then

$$\sigma\tau = (13), \quad \tau\sigma = (23),$$

so

$$\sigma H = \{\sigma, (13)\} \neq \{\sigma, (23)\} = H\sigma.$$

Therefore H is not normal in S_3 , even though $S_3/A_3 \cong H \cong \mathbb{Z}_2$. Isomorphism does not identify a quotient group with a particular subset of the original group.

What to retain from this chapter

1. A subgroup $N \leq G$ is normal exactly when $gN = Ng$, equivalently $gNg^{-1} = N$, for every $g \in G$.
2. Coset multiplication $(aN)(bN) = (ab)N$ is well defined exactly when $N \trianglelefteq G$; it then makes G/N a group.
3. Kernels are normal, and the image of a normal subgroup is normal in the image of the homomorphism.
4. The canonical projection $G \rightarrow G/N, a \mapsto aN$, is surjective and has kernel N .
5. Every subgroup of an Abelian group is normal, and $Z(G) \trianglelefteq G$.
6. A simple group has no nontrivial proper normal subgroup. The cyclic group $\mathbb{Z}_n$ is simple exactly when n is prime, while A_4 is not simple because $V_4 \trianglelefteq A_4$.
7. $A_n \trianglelefteq S_n$ and $S_n/A_n \cong \mathbb{Z}_2$, but a quotient group is not in general a subgroup of its parent group.